

# When Does LLM-Serving Scheduler Adaptation Matter? Action Opportunity and Causal Headroom in Production-Derived Replay

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## Abstract

Adaptive scheduling for large language model (LLM) serving pays off only if the live system reaches states where a different executable action exists and taking it improves performance. We measure it before any adaptive scheduler is designed. In a deterministic simulator replaying production-derived traces (Azure code, Azure conversation, BurstGPT), we count how often a six-policy portfolio proposes an action different from the single best scheduler’s and, on fresh non-overlapping windows, force one alternative action for one step before returning to the default. Under abundant resources, replay was action-null in about one million decision states. Capping key-value (KV) cache capacity or concurrent sequences created disagreement; scaling arrival rates up to eight times did not. In a preregistered analysis of 720 fresh disagreement states, 590 (81.9%) had positive one-step latency headroom; the state-weighted mean oracle headroom, the best available one-step latency reduction, was 2.00 ms (95% window-clustered confidence interval 0.19–3.55 ms). Post hoc analyses show that the sign is stable across weightings and leave-one-out analyses, but the magnitude is not: the median is 0.20 ms, and one of 36 windows holds 88% of the total headroom. A bounded vLLM probe shows that resource pressure changes scheduler behavior but does not validate the simulated latency effect. Adaptation opportunity is real but sparse, pressure-dependent, and generally modest; detecting it does not show that a deployed adaptive scheduler is worthwhile.

*Keywords:* LLM serving, request scheduling, performance evaluation, workload replay, resource contention, causal evaluation

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## 1. Introduction

Adaptive scheduling in large language model (LLM) serving is usually motivated by heterogeneous prompts, decode demand, key-value (KV) cache residency, queueing, and deadlines. It can help only if the live system reaches states where a different executable action exists, such as another admission, batch, or request order, and only if taking that action improves the target metric by more than the cost of choosing it. How often this happens is rarely measured before a scheduler is designed.

Common evaluation practice does not answer this. Ranking policies on whole-trace metrics, or reporting the gap to a virtual best solver (VBS) as in algorithm portfolios [1, 2], shows that policies differ but aggregates thousands of decisions; it cannot say whether an alternative was available at a given state. Policies with different scores can issue the same batch, and an alternative that exists may not reduce latency. Recent systems address *how* to adapt [3, 4, 5, 6]. We ask the prior question: *when is there a different useful action to take, and how much is it worth?*

We answer it with a staged decomposition,

$$\begin{aligned} \text{resource pressure} &\longrightarrow P(D) \longrightarrow P(B_{\text{LAT}} \mid D) \\ &\longrightarrow \text{headroom magnitude} \longrightarrow \text{predictability and deployment.} \end{aligned} \tag{1}$$

Here  $D$  means that at least one canonical executable alternative differs from the default single best scheduler, and  $B_{\text{LAT}}$  that at least one such alternative lowers one-step mean latency relative to that default. These are distinct quantities: pressure can make policies disagree often without the disagreement being useful, and useful disagreement can be rare or small. The last arrow, whether a selector could exploit the opportunity, is not measured here.

This is a performance-evaluation question: at which operating points does scheduler choice have causal performance consequences? Answering it takes workload characterization, controlled resource interventions, replay-based simulation, and inference that respects the dependence among states from one trace window. Recent work in *Performance Evaluation* analyzes job assignment in machine-learning inference systems under accuracy constraints [7] and adaptive machine-learning training on serverless clouds [8]; our measurement-first view characterizes the region in which such mechanisms have something to choose between.

We replay production-derived traces (Azure 2023 code and conversation [9], BurstGPT [10]) in a deterministic simulator. Faithful replay of 60 windows gives the native baseline; one-axis interventions on active-sequence capacity, KV capacity, and arrival rate (1,080 conditions) locate where disagreement appears. On fresh non-overlapping windows we clone the simulator at each disagreement state, force one alternative action once, and return to the default. The primary endpoint, oracle latency headroom (the largest one-step mean-latency reduction among the alternatives, zero if none helps), was fixed in a preregistered protocol; the robustness analyses of the frozen results are post hoc. A bounded vLLM [11] probe checks the mechanism in a real engine.

Native replay is action-null under abundant resources. Capacity caps on active sequences or KV cache create disagreement; arrival-rate scaling up to eight times does not. Of 720 fresh disagreement states, 590 (81.9%; 589 excluding one floating-point-residue state) had positive headroom, with a state-weighted mean of 2.00 ms (95% window-clustered confidence interval 0.19–3.55 ms). The sign is robust: the mean stays positive under equal-window, equal-regime, and equal-workload weighting and when any single window or regime is removed. The magnitude is not uniform: the median is 0.20 ms, only 22.5% of states offer more than 2 ms, and one of 36 windows carries 88% of the total headroom. The vLLM probe shows that pressure alters queueing and scheduling behavior but does not validate the simulator’s latency magnitudes. Adaptation opportunity is thus real but sparse, pressure-dependent, and generally modest, and detecting it does not show that a deployed scheduler is worthwhile. The evidence covers three trace families, a six-policy portfolio, a simulator, and one bounded probe on a single model and GPU; the oracle is an opportunity bound, not a scheduler.

The study addresses five research questions, ordered along the decomposition.

**RQ1 – Native opportunity.** How often does the single best scheduler face a different executable action from the six-policy portfolio, in faithful replay of production-derived traces with abundant resources?

**RQ2 – Pressure-induced opportunity.** Which resource-pressure dimensions create action disagreement, and does the transition reproduce on untouched windows?

**RQ3 – Causal value.** At disagreement states, how often does forcing one

alternative action for one step reduce latency, and does the answer depend on the objective?

**RQ4 – Magnitude and concentration.** How large is the headroom, is it typical or concentrated, and how sensitive is it to weighting and to individual windows or regimes?

**RQ5 – System correspondence.** Does a bounded real vLLM deployment show the corresponding pressure and scheduling mechanism?

### *1.1. Contributions*

This paper makes five contributions.

1. A staged action-opportunity evaluation that separates executable disagreement, beneficial disagreement, and headroom magnitude from policy-score diversity, with prevalence measured over all valid decision states and uncertainty quantified by trace window.
2. A characterization of where disagreement appears: native replay is action-null, and only capacity limits on active sequences or KV cache, not arrival-rate scaling, expose alternatives; the transition reproduces on untouched windows.
3. A preregistered one-step causal measurement on 720 fresh states, with latency as the primary endpoint because deadline goodput saturates at 1.0 and hides latency differences.
4. A robustness and concentration analysis showing that the sign of the headroom is robust but its magnitude is concentrated, so a mean reported without its distribution and cluster composition overstates the typical opportunity.
5. Bounded system evidence that resource pressure changes queueing and scheduling behavior in vLLM, without validating simulated latency magnitudes or a deployed selector.

Section 2 positions the work, and Section 3 defines the estimands and evidence design. Sections 4–6 report the native, pressure, and fresh causal results, Section 7 tests their robustness and concentration, Sections 8 and 9 cover objective sensitivity and the vLLM probe, and Section 10 discusses when adaptation is justified.

## 2. Related Work and Positioning

This section positions the paper against LLM-serving systems, workload studies, and algorithm-selection work. LLM-serving research spans model execution, prefill/decode scheduling, resource management, routing, caching, and cluster control. Orca and vLLM established iteration-level serving and paged KV management [12, 11]. Sarathi-Serve, DistServe, Splitwise, and FastServe study chunking, phase separation, and preemptive scheduling [13, 5, 14, 15]. Llumnix and SOLA adapt scheduling or state-aware service to dynamic conditions [3, 4]; Libra studies flexible request partitioning [16]; Mooncake and Strata emphasize KV-centric or hierarchical context caching [17, 18]; MorphServe changes model precision and KV capacity under pressure [19]; and DynamoLLM reconfigures cluster resources for performance and energy [20]. BurstGPT and Azure traces provide important workload evidence [10, 9]; the recent longitudinal study by Nixon et al. further emphasizes workload evolution, caching, and load-balancing over a year of production traffic [21].

ServeGen and OpenTela narrow any generic claim that workload realism or operational trace infrastructure is itself novel [22, 23]. LMetric is especially close: it combines a KV-aware indicator with an instance-load indicator, reports production-style workloads, and derives conditions under which its multiplication can fail [6]. These works answer how to build or evaluate better serving mechanisms. Our question is complementary: in a fixed workload/resource regime, do the default and alternatives expose different canonical actions at all, and does forcing one alternative action have default-relative causal value?

For this reason, systems such as ServeGen, Libra, Llumnix, MorphServe, Strata, LMetric, and OpenTela are related work and design context, not unimplemented baselines for a superiority claim. Reproducing each system would be necessary for a paper claiming that a new scheduler beats them. This paper instead asks whether an executable action opportunity and local default-relative headroom exist before building an adaptive selector. The single best scheduler/VBS-style quantities used below are non-deployable references for opportunity measurement, not schedulers proposed for production.

Table 1 records the scope honestly. A “partial” entry means that a work contains related evidence but does not use the paper’s canonical action or one-step estimand.

Our use of single best scheduler/VBS is rooted in algorithm portfolios

Table 1: Closest-work differentiation. We do not claim that other systems omit pressure, production-derived traces, or adaptive mechanisms; the distinction is the action-level and default-relative causal measurement question.

| Work                 | Prod.-<br>derived<br>workloads | Pressure<br>sweep | Proposes<br>system | Canonical<br>disagree-<br>ment | One-step<br>default in-<br>tervention | Real<br>system            |
|----------------------|--------------------------------|-------------------|--------------------|--------------------------------|---------------------------------------|---------------------------|
| ServeGen             | Yes                            | No                | Generator          | No                             | No                                    | Workload<br>study         |
| LMetric              | Yes                            | Implicit          | Yes                | No                             | No                                    | Yes                       |
| OpenTela             | Operational<br>traces          | No                | Control<br>plane   | No                             | No                                    | Yes                       |
| Llumnix /<br>SOLA    | Selected<br>regimes            | Partial           | Yes                | No                             | No                                    | System<br>evalua-<br>tion |
| Mooncake<br>/ Strata | Cache/resource<br>regimes      | Partial           | Yes                | No                             | No                                    | Yes /<br>opera-<br>tional |
| This work            | Yes                            | Yes               | No                 | Yes, $P(D)$                    | Yes,<br>simulator<br>only             | Bounded<br>partial        |

and selection [1, 2, 24, 25]. The novelty claim is not the envelope itself, nor counterfactual reasoning in general. It is the narrow serving characterization in Eq. 1.

### 3. Methods

#### 3.1. Evidence hierarchy and workload inventory

We report four evidence tiers. Tier 1 is native production-derived trace replay. Tier 2 is production-derived aggregate or window validation. Tier 3 is a controlled transformation of a real trace, such as one-axis resource caps. Tier 4 is synthetic or mechanistic workload construction. RQ1 uses Tier 1; RQ2 uses Tiers 2–3; fresh causal headroom uses a preregistered Tier-3 intervention over untouched windows; the real-vLLM result is a bounded system validation rather than a claim of production deployment.

The faithful corpus contains 20 windows each from Azure 2023 code, Azure 2023 conversation, and BurstGPT. Source order and token-length/arrival structure are retained. SLO and class overlays are deterministic simulator inputs; future actual output lengths are hidden from policies. The fresh confirmatory campaign uses 20 untouched windows per workload family for support mapping, with causal labeling limited to the frozen selected regimes. BurstGPT did not contribute a selected fresh causal disagreement regime, which is retained as an explicit heterogeneity limitation.

The historical six-policy portfolio contains full prefill, small-chunk prefill, estimated-service-first (ESTF), weighted fair sharing (WFS), least-laxity first (LLF), and a KV-constrained policy. The six-policy portfolio is fixed for a default-relative characterization; it is not an exhaustive set of 2026 schedulers.

### 3.2. Canonical action opportunity

Let  $\mathcal{A}_6(s)$  be the set of canonical executable actions produced by the six portfolio policies at state  $s$ , after action deduplication. A canonical action is the simulator-executable admission, batch, and request ordering decision after infeasible or semantically identical variants have been collapsed. The default scheduler is the single best scheduler (SBS), the fixed best single policy used as the canonical baseline for the frozen replay protocol; it is not refit per state. Let  $a_{\text{SBS}}(s)$  be its action. We define

$$D(s) = \mathbf{1} [\exists a \in \mathcal{A}_6(s) : a \neq a_{\text{SBS}}(s)] . \quad (2)$$

The opportunity prevalence is  $P(D) = \sum_s D(s) / |\mathcal{S}_{\text{all}}|$ , where the denominator is every valid single best scheduler decision state, not only states where policy scores differ. A policy ranking change that produces the same admission, batch, and request order is action collapse, not opportunity. An alternative action is any executable member of  $\mathcal{A}_6(s)$  that is not equal to  $a_{\text{SBS}}(s)$ .

This definition matters operationally. Whole-policy comparisons can show a latency or goodput difference accumulated over many decisions, but cannot say whether an alternative existed at a particular default state. Conversely, action diversity alone says nothing about usefulness.

### 3.3. One-step causal headroom

For every disagreement state  $s$  and canonical alternative  $a$ , we clone the state, force exactly one action, and return to single best scheduler for the

continuation. Denote the resulting continuation value by  $Q_{\text{SBS}}(s, a)$ . This is a local counterfactual, not selector performance: it estimates the one-step effect of changing the current action while holding the subsequent default policy fixed.

For the fresh primary latency endpoint, continuation mean latency includes active, queued, and future requests belonging to the continuation, excluding requests completed before  $s$ . We define

$$A_{\text{LAT}}(s, a) = L_{\text{SBS}}(s) - L_{\text{CF}}(s, a), \quad H_{\text{LAT}}(s) = \max \left( 0, \max_a A_{\text{LAT}}(s, a) \right), \quad (3)$$

where positive values reduce mean latency. The beneficial-state indicator is  $B_{\text{LAT}}(s) = 1$  iff  $\max_a A_{\text{LAT}}(s, a) > 0$ . We report  $P(B_{\text{LAT}} \mid D)$  and mean  $H_{\text{LAT}}$  separately from  $P(D)$ . The conceptual aggregate opportunity is therefore

$$P(D) \times P(B_{\text{LAT}} \mid D) \times \text{effect magnitude}, \quad (4)$$

not an exact deployable-performance equation. A high conditional benefit can coexist with a tiny overall opportunity.

The confirmatory mean uses a faithful-window clustered percentile bootstrap with 2,000 replicates and seed 20260920. The cluster unit is the original fresh trace window, identified by workload source and window index, contributing disagreement states, so multiple states from one window do not become independent experimental units. Its verdict requires positive mean headroom and a clustered 95% lower endpoint above zero.

A resource-pressure regime is a workload window plus one frozen one-axis intervention, such as a KV-capacity cap, active-sequence cap, or arrival-rate scaling factor. “Support” and “opportunity” refer only to the existence of executable non-single best scheduler actions under that regime; usefulness requires the separate causal latency test above.

### 3.4. *Post hoc robustness analyses*

The analyses in Section 7 were specified after the fresh outcomes were known. They qualify the preregistered result and do not replace it: the endpoint  $H_{\text{LAT}}$ , the strict criterion  $B_{\text{LAT}}$ , the 720 states, and the selected regimes are unchanged. A *window* is a workload source and window index (36 windows contribute states); a *regime* is a workload, pressure axis, and pressure setting (five selected regimes).

*Practical thresholds.* For  $\epsilon \in \{0.1, 0.25, 0.5, 1, 2, 5\}$  ms we report the number and share of disagreement states with  $H_{\text{LAT}}(s) > \epsilon$ , that is  $P(H_{\text{LAT}} > \epsilon \mid D)$ , and the number of windows that contain at least one such state.

*Numerical residue.* The preregistered criterion is applied unchanged in double precision. Simulated latency differences are integer multiples of the decode step divided by the request population, so no genuine advantage is smaller than about  $5.2 \times 10^{-6}$  s. Only to identify floating-point residue, we also count states whose best advantage exceeds  $10^{-12}$  s.

*Aggregation.* For a unit  $u$  (a window, regime, or workload source) let  $D_u$  be its disagreement states and  $\bar{H}_u = |D_u|^{-1} \sum_{s \in D_u} H_{\text{LAT}}(s)$ . The preregistered estimate is the *state-weighted mean*  $\bar{H} = |D|^{-1} \sum_{s \in D} H_{\text{LAT}}(s)$ , in which every state has weight  $1/|D|$ . The *equal-window*, *equal-regime*, and *equal-workload* means are  $U^{-1} \sum_u \bar{H}_u$  over the 36 windows, five regimes, or two workloads, so every unit has weight  $1/U$  however many states it contains. These are different estimands, not alternative estimates of one quantity; the beneficial share is aggregated in the same way. Intervals for the alternative means use the same window-clustered percentile bootstrap with 200,000 replicates. For the equal-regime and equal-workload means, regimes and workloads are fixed strata and window–regime cells are resampled within each stratum, which ignores dependence between regimes that share a window.

*Leave-one-out and concentration.* For each window (36) and each regime (5) we recompute the state-weighted mean and beneficial share from the remaining states, with a window-clustered bootstrap interval (200,000 replicates). Concentration is the share of the total summed headroom  $\sum_s H_{\text{LAT}}(s)$  contributed by each window and regime, and the effective number of windows  $1/\sum_w p_w^2$ , where  $p_w$  is the share of window  $w$ .

*Interval diagnostics.* To check the preregistered interval we repeat the percentile bootstrap with  $10^6$  replicates, add a bias-corrected and accelerated (BCa) interval, and compute a delete-one-window jackknife  $t$  interval. These are diagnostics, not replacements.

### 3.5. Objective audit

The historical objective is arrival-normalized weighted goodput (ANWG),

$$\text{ANWG} = \frac{\sum_i w_i \mathbf{1}[i \text{ completes by its SLO}]}{\sum_i w_i}. \quad (5)$$

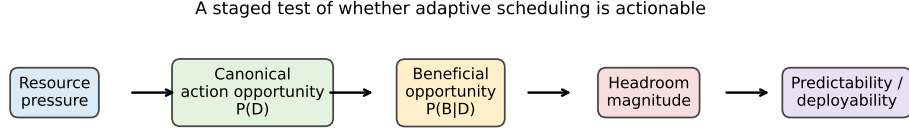


Figure 1: The paper’s evidence chain. A predictor is downstream of both action opportunity and causal value; deployment benefit is not assumed by the characterization.

The earlier terminal arrival-normalized weighted goodput result is retained as an objective audit, not as fresh latency evidence. The latency study uses Eq. 3; its mean-latency result is confirmatory only for the fresh untouched causal universe.

#### 4. Results: Native Action Opportunity

Under the frozen abundant-resource baseline, all three production-derived workloads were canonically action-null (Figure 1 summarizes the evidence chain; Figure 2 gives the rates). Azure code had 0 disagreements in 99,992 single best scheduler decision states; Azure conversation had 0/461,985; and BurstGPT had 0/440,461. The corresponding maximum KV utilizations were 0.002716, 0.001728, and 0.003802, with no active-cap binding states. Policy-ranking proxy differences existed in a small number of states, but canonical actions collapsed. This is not evidence that adaptive scheduling never matters; it is evidence that this resource regime offered no different executable six-policy action.

#### 5. Results: Resource-Regime Transition

The Resource-Regime map evaluated 1,080 one-axis conditions: 20 windows across three workload families, with active-sequence, KV-capacity, and arrival pressure axes (Figure 3; Table 2). There were 926 valid unconstrained, 38 partially constrained, 96 strongly constrained, and 20 horizon-invalid conditions. Across valid conditions, 11,328 canonical disagreement states were observed.

Arrival scaling through  $8\times$  remained action-null for all three workload families. Active-sequence pressure created disagreement at active-cap 8 or 4 depending on workload; KV pressure created disagreement at the first

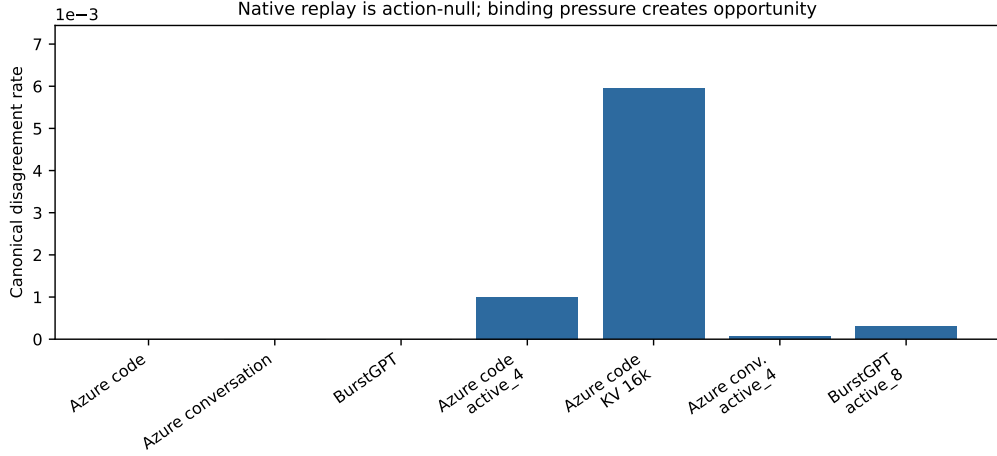


Figure 2: Native production-derived replay is action-null. Selected pressure conditions create nonzero canonical disagreement, with rates depending on workload and pressure axis. This is support for adaptive action selection, not evidence of selector benefit by itself.

binding or near-binding capacities. For Azure code, the first active-cap disagreement appeared at active-sequence cap = 8 and became sustained at active-sequence cap = 4. KV disagreement began at KV capacity = 16,000. Azure conversation first disagreed at active-sequence cap = 4 and showed KV disagreement at KV capacity = 16,000; BurstGPT showed active disagreement at active-sequence cap = 16, sustained at active-sequence cap = 8, and KV disagreement at KV capacity = 16,000.

The transition (Table 2) is the main regime result: pressure does not automatically create useful diversity, but concrete resource binding can turn a collapsed action set into a nontrivial one. In the studied traces, arrival-rate scaling alone did not create support through the tested range; tight KV capacity and active-sequence limits were the operative dimensions.

## 6. Results: Fresh Support and Causal Latency Headroom

The fresh support mapping used untouched windows and the frozen pressure protocol. The causal labeling plan selected five Azure code/conversation active/KV regimes and 720 disagreement states. BurstGPT contributed no selected causal regime, an important limitation rather than a missing result.

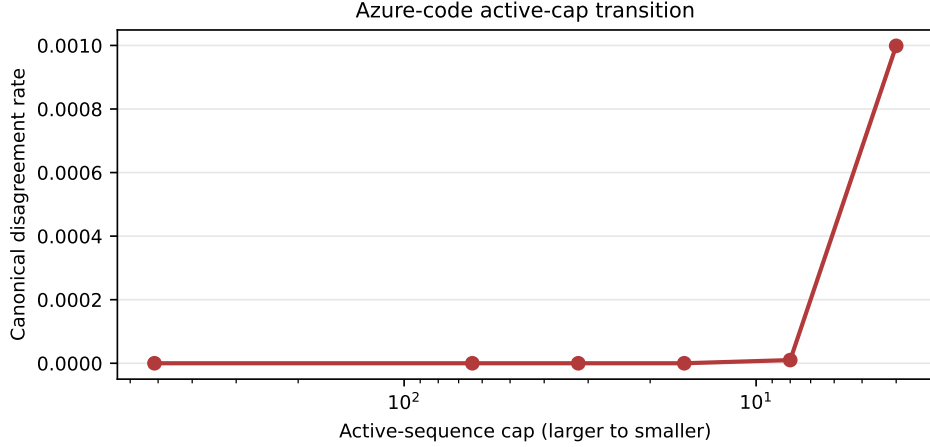


Figure 3: Azure-code active-sequence transition. Agreement persists until the active cap binds the trace, after which canonical disagreement increases.

The confirmatory intervention found 590 beneficial states (Table 3; Figure 4), giving

$$P(B_{\text{LAT}} \mid D) = 590/720 = 0.8194444444.$$

Mean oracle latency headroom was 0.001995791 s (1.9958 ms), with a clustered 95% CI of [0.000190884, 0.003546196] s ([0.1909, 3.5462] ms). The preregistered verdict was positive. The result is statistically supported but operationally modest. It establishes measurable local default-relative latency opportunity in the disagreement-support regime, not that the magnitude exceeds the engineering cost of adaptation. It is also not the claim that 81.9% of all scheduler decisions can be improved:  $P(D)$  is often below one percent in the selected regimes and the conditional statistic is defined only on disagreement states.

These preregistered values are state-weighted over disagreement states drawn from 36 windows, and they should be read with the post hoc analyses of Section 7: the typical state has a much smaller headroom (median 0.20 ms), and the mean is highly concentrated in one window. One of the 590 beneficial states has an advantage of  $5.6 \times 10^{-17}$  s, exactly one unit in the last place of the 0.357 s latencies compared, with both branches completing the same requests; a post hoc guard of  $10^{-12}$  s gives 589/720 (81.8%) and changes the mean by less than  $10^{-19}$  s.

Table 2: Pressure-to-disagreement transition summary. “Sustained” means at least two windows with disagreement.

| Workload           | Axis       | First binding               | First disagreement          | Sustained                  |
|--------------------|------------|-----------------------------|-----------------------------|----------------------------|
| Azure code         | Active cap | active-sequence cap<br>= 8  | active-sequence cap<br>= 8  | active-sequence cap<br>= 4 |
| Azure code         | KV cap     | KV capacity<br>= 16,000     | KV capacity<br>= 16,000     | KV capacity<br>= 16,000    |
| Azure conversation | Active cap | active-sequence cap<br>= 4  | active-sequence cap<br>= 4  | active-sequence cap<br>= 4 |
| Azure conversation | KV cap     | KV capacity<br>= 8,000      | KV capacity<br>= 16,000     | KV capacity<br>= 8,000     |
| BurstGPT           | Active cap | active-sequence cap<br>= 16 | active-sequence cap<br>= 16 | active-sequence cap<br>= 8 |
| BurstGPT           | KV cap     | KV capacity<br>= 16,000     | KV capacity<br>= 16,000     | KV capacity<br>= 16,000    |

### 6.1. Opportunity decomposition

Figure 5 makes the distinction between prevalence and value concrete. Azure-code KV capacity = 16,000 has the largest selected  $P(D)$  and the largest mean headroom; active-cap regimes have much smaller  $P(D)$  and correspondingly smaller aggregate opportunity even when some disagreement states are beneficial. A practitioner should therefore measure all three factors in Eq. 4: whether a different action exists, whether it helps conditional on existing, and whether its effect is large enough to matter relative to system noise and control overhead.

The practical reading is deliberately staged. Under abundant resources, this study observed no canonical disagreement and therefore no demonstrated action support for adaptive selection. Under constrained KV or active-sequence regimes, disagreement emerges and creates an executable opportunity. Within those disagreement states, alternatives with positive headroom are frequent but mostly small (Section 7). Opportunity and headroom still

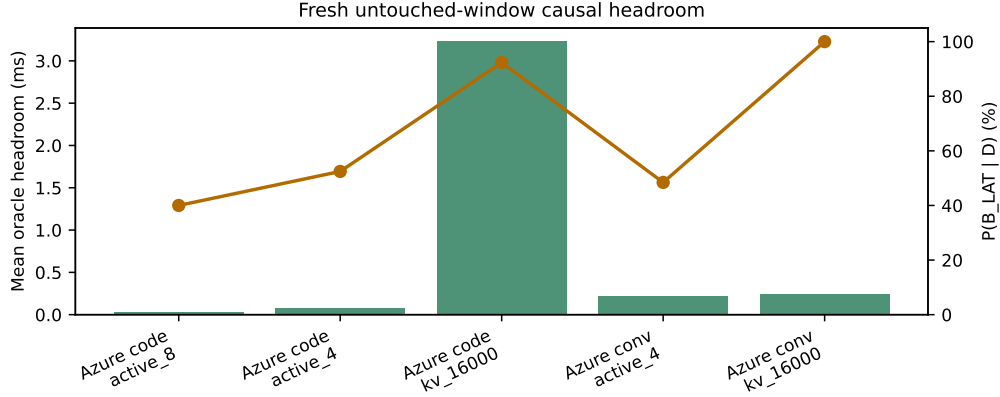


Figure 4: Fresh causal result by selected workload/regime. Bars show mean oracle headroom in milliseconds; the line shows  $P(B_{LAT} | D)$ . The Azure-code KV-capacity = 16,000 regime holds 59.9% of the 720 states and 96.8% of the total headroom. The oracle is a non-deployable opportunity bound over observed alternatives.

do not prove that a deployed adaptive selector is worthwhile.

## 7. Results: Robustness and Concentration of Causal Headroom

The preregistered result of Section 6 is a state-weighted mean over 720 disagreement states drawn from 36 windows. This section asks, post hoc, how large the headroom is in a typical state and how far the mean depends on which windows and regimes contribute (Section 3.4). The preregistered criterion and estimate are not modified.

### 7.1. Effect scale and distribution

Table 4 counts the states whose headroom exceeds each practical threshold. The 81.9% beneficial share is not the share of states that offer material gains: 60.0% of states exceed 0.1 ms, 29.4% exceed 0.5 ms, 22.5% exceed 2 ms, and 17.9% exceed 5 ms. The larger effects are also concentrated. Above 0.5 ms, 201 of the 212 states belong to the Azure-code KV-capacity = 16,000 regime, and only 4 of the 36 windows contain a state above 2 ms.

The distribution is strongly right-skewed (Figure 6a). The median headroom is 0.197 ms, about one tenth of the 1.996 ms state-weighted mean, and 77.4% of states lie at or below that mean. The quartiles are 0.023 and

Table 3: Fresh causal characterization.  $P(D)$  uses all valid single best scheduler decision states;  $P(B_{\text{LAT}} | D)$  uses only the disagreement states. Per-regime source-window counts report distinct fresh trace windows with at least one disagreement state in that regime. The same source window may contribute to multiple regimes, so the row counts are not additive; the 36 distinct workload-window pairs form the clusters used by the bootstrap.

| Workload/regime                             | States | $P(D)$   | $P(B_{\text{LAT}}   D)$ | Mean $H_{\text{LAT}}$ (ms) | Source windows |
|---------------------------------------------|--------|----------|-------------------------|----------------------------|----------------|
| Azure code, active-sequence cap = 8         | 10     | 0.000106 | 0.400                   | 0.033                      | 2              |
| Azure code, active-sequence cap = 4         | 122    | 0.001290 | 0.525                   | 0.077                      | 15             |
| Azure code, KV capacity = 16,000            | 431    | 0.004561 | 0.923                   | 3.227                      | 14             |
| Azure conversation, active-sequence cap = 4 | 64     | 0.000108 | 0.484                   | 0.217                      | 17             |
| Azure conversation, KV capacity = 16,000    | 93     | 0.000157 | 1.000                   | 0.241                      | 1              |
| All selected fresh states                   | 720    | —        | 0.819                   | 1.996                      | 36 distinct    |

1.19 ms, the 90th and 95th percentiles are 8.4 and 11.3 ms, and the maximum is 12.1 ms. A further 130 states (18.1%) have exactly zero headroom. Among the 589 states with positive headroom the median is 0.29 ms.

### 7.2. Sensitivity to aggregation and to individual windows and regimes

Every aggregation of Table 5 yields a positive mean whose interval excludes zero, but the magnitude depends on what is weighted: from 1.996 ms (state-weighted) to 0.327 ms (equal-window), a factor of about six. The beneficial share is likewise composition-sensitive, from 0.819 (state-weighted) to 0.632 (equal-window). The state-weighted mean is the preregistered estimand and answers “what is the expected headroom at a disagreement state drawn from this sample”; the equal-window and equal-regime means answer the same question for a typical window or regime, and they are not interchangeable with it.

Leaving one unit out gives the same picture. All 36 single-window and

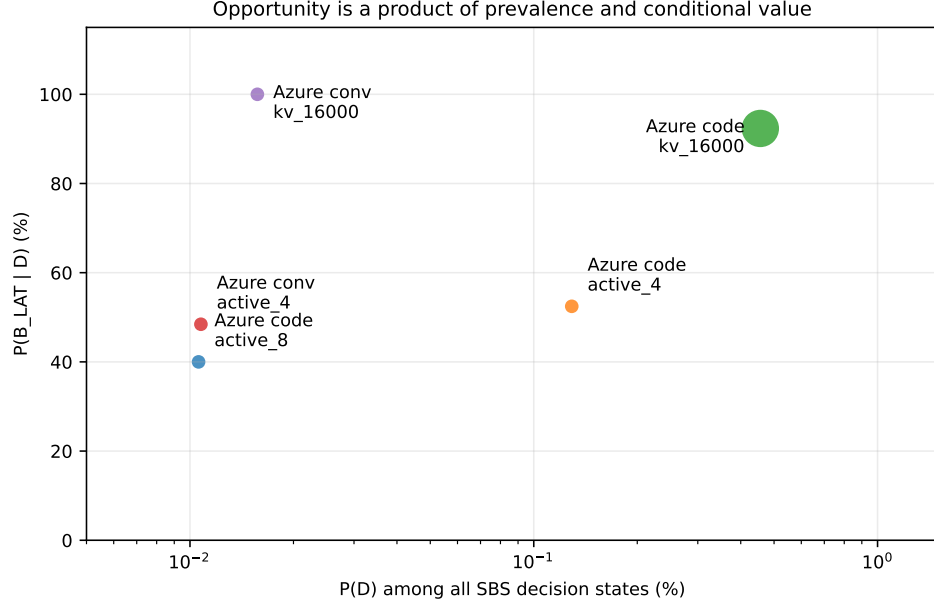


Figure 5: Regime map of opportunity prevalence  $P(D)$  (over single best scheduler decision states) against conditional benefit  $P(B_{LAT} | D)$  for the five selected regimes; point size is proportional to the regime’s mean oracle headroom.  $P(B_{LAT} | D)$  counts any positive headroom, however small: the regime at 100% (Azure conversation, KV-capacity = 16,000) never exceeds 0.48 ms of headroom.

5 single-regime omissions leave a positive mean, and every corresponding bootstrap interval excludes zero. Omitting window w11 leaves 0.365 ms (95% CI [0.158, 0.682] ms); omitting the Azure-code KV-capacity = 16,000 regime leaves 0.159 ms (95% CI [0.058, 0.234] ms). Omitting any other window changes the mean only to between 2.00 and 2.26 ms, and omitting any other regime to between 2.02 and 2.39 ms. The sign therefore survives all 41 omissions, whereas the headline magnitude does not survive the omission of the two dominant units.

### 7.3. Concentration of headroom

Window w11 of the Azure-code trace contains 263 of the 720 states (36.5%) but contributes 88.4% of the total summed headroom; the three largest windows contribute 95.3%, and the effective number of windows by headroom mass is 1.27 (5.7 by state count). At regime level, Azure-code KV-capacity = 16,000 holds 59.9% of the states and 96.8% of the headroom,

Table 4: Post hoc practical-effect thresholds: disagreement states whose oracle headroom  $H_{\text{LAT}}$  exceeds each threshold. The first row is the preregistered criterion; the second excludes one floating-point-residue state with a  $10^{-12}$  s guard. “Share” is of the 720 states; “Windows” counts the windows (of 36) that contain at least one such state; “In KV 16,000” counts states from the Azure-code KV-capacity = 16,000 regime, which contributes 431 of the 720 states.

| Criterion                       | States | Share | Windows | In KV 16,000 |
|---------------------------------|--------|-------|---------|--------------|
| Preregistered ( $> 0$ , strict) | 590    | 81.9% | 32      | 398          |
| Guarded ( $> 10^{-12}$ s)       | 589    | 81.8% | 32      | 398          |
| $> 0.1$ ms                      | 432    | 60.0% | 17      | 327          |
| $> 0.25$ ms                     | 322    | 44.7% | 16      | 259          |
| $> 0.5$ ms                      | 212    | 29.4% | 9       | 201          |
| $> 1$ ms                        | 188    | 26.1% | 7       | 185          |
| $> 2$ ms                        | 162    | 22.5% | 4       | 161          |
| $> 5$ ms                        | 129    | 17.9% | 2       | 128          |

contributing 1.93 of the 2.00 ms state-weighted mean. Within w11, 205 states of that regime average 6.16 ms. Of the 129 states above 5 ms, 128 are in w11, as are 140 of the 162 states above 2 ms (Figure 6b). The state-weighted mean is thus highly concentrated in a narrow high-headroom subset of states and should not be read as the typical benefit of a disagreement state.

#### 7.4. Uncertainty diagnostics

Monte Carlo error does not explain the width of the preregistered interval. With  $10^6$  replicates the percentile interval is  $[0.184, 3.543]$  ms, no replicate has a nonpositive mean, and the lower endpoint of the 2,000-replicate procedure varies by only 0.004 ms (standard deviation) across 1000 seeds. The BCa interval is  $[0.239, 4.049]$  ms. The bootstrap distribution is bimodal: 36% of replicates do not draw window w11 and have a mean of about 0.36 ms, against 2.43 ms for those that do, so the percentile interval spans the gap between two populations of replicates. The delete-one-window jackknife  $t$  interval,  $[-1.36, 5.35]$  ms, includes zero because one influential window inflates its standard error (1.65 ms); all 36 delete-one estimates are positive, and we treat the jackknife interval as a diagnostic of this dependence rather than as a competing interval. Uncertainty about the magnitude of the headroom thus reflects its concentration in few windows, not simulation noise.

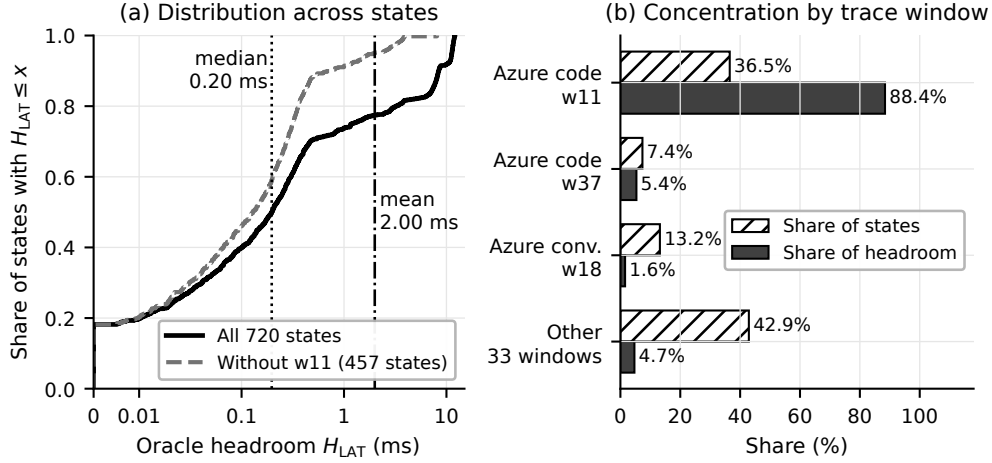


Figure 6: Distribution and concentration of oracle headroom over the 720 disagreement states. (a) Empirical distribution of  $H_{LAT}$  with and without window w11; zero headroom is the intercept at  $x = 0$ , and the axis is linear below 0.01 ms and logarithmic above. The dotted and dash-dotted lines mark the median and the state-weighted mean. (b) Share of the 720 states (hatched) and of the total summed headroom (solid) for the three windows with the largest headroom and for the remaining 33 windows.

## 8. Objective Sensitivity: Arrival-Normalized Weighted Goodput Saturation

The earlier pressure study completed with causal integrity and complete branch coverage, but all terminal ANWG values were 1.0 and all ANWG advantages were zero. An identifiability audit found no implementation bug: requests completed within deadlines and the objective saturated. Simple sensitivity cases confirmed that ANWG distinguishes missed deadlines or unfinished work, but not different latency or queueing when all requests still meet a loose deadline.

Most counterfactual branches nevertheless changed latency. Thus zero ANWG headroom does not imply zero latency headroom. We do not use the earlier post hoc latency analysis as confirmatory evidence. The fresh untouched study in Section 6 is the confirmatory latency result, with its own pre-frozen primary endpoint and clustered bootstrap.

This is objective sensitivity, not evidence that ANWG is invalid. It is an appropriate deadline-goodput objective for one question; mean latency is a different operational objective. A serving paper should not promote

Table 5: Post hoc sensitivity of the mean oracle headroom to the aggregation rule and to leaving out the two dominant units. In the first block each workload, regime, or window receives equal weight, except in the state-weighted row, where each state does. “Share” is the beneficial share under the strict  $> 0$  criterion. The state-weighted row is the pre-registered estimate and interval (2,000 replicates); all other intervals are window-clustered percentile bootstrap intervals with 200,000 replicates (equal-regime and equal-workload treat regimes and workloads as fixed strata; Section 3.4).

| Estimand                                                          | Units       | Mean (ms) | 95% CI (ms)    | Share |
|-------------------------------------------------------------------|-------------|-----------|----------------|-------|
| State-weighted (preregistered)                                    | 720 states  | 1.996     | [0.191, 3.546] | 0.819 |
| Equal-workload                                                    | 2 workloads | 1.360     | [0.184, 2.094] | 0.809 |
| Equal-regime                                                      | 5 regimes   | 0.759     | [0.145, 1.161] | 0.666 |
| Equal-window                                                      | 36 windows  | 0.327     | [0.114, 0.644] | 0.632 |
| <i>Leave one out: state-weighted mean of the remaining states</i> |             |           |                |       |
| Omit window w11                                                   | 457 states  | 0.365     | [0.158, 0.682] | 0.821 |
| Omit Azure-code KV 16,000                                         | 289 states  | 0.159     | [0.058, 0.234] | 0.664 |

whichever metric yields the more favorable conclusion.

## 9. Bounded Real-vLLM Validation

The bounded real-vLLM validation used one RTX 5060 Ti, vLLM 0.27.1, Qwen2.5-0.5B-Instruct, and deterministic token-shape replay. It completed 40 full/chunked runs and a 20-run native chunked-prefill budget probe. Queueing and scheduler traces were retained. The verdict partially supported means support for the mechanism but not for the full causal or deployment claim. Level 1 is demonstrated only in controlled replay/intervention: at disagreement states, changing the canonical action can expose one-step latency headroom. Level 2 is supported in real vLLM: resource pressure changes observable queue and scheduling behavior. Level 3 is not demonstrated: the paper does not show that a deployed adaptive selector on real vLLM would recover 1.9958 ms or improve end-to-end latency, SLOs, or throughput.

The native budget probe held chunked prefill fixed and varied the token budget. T512 produced 1,680 scheduler steps, 165 mixed prefill/decode steps, and 194 partial-prefill items. T4096 produced 1,536, 55, and 9. Waiting queues reached 7 and 5, respectively, while running sequences reached 4 in both cases. The low-load regime showed a 30.6 ms late-time-to-first-token

Table 6: Simulator-to-real-system correspondence. The real-system column supports pressure/action correspondence, not one-step causal headroom.

| Property            | Simulator/replay                                  | Real vLLM                                   |
|---------------------|---------------------------------------------------|---------------------------------------------|
| Low pressure        | Canonical collapse                                | Low queue baseline                          |
| Resource binding    | KV/active caps induce $D$                         | Queue and active-sequence pressure observed |
| Action difference   | single best scheduler/six-policy canonical action | Native token-budget trace signature         |
| Latency consequence | One-step single best scheduler-relative           | Whole-run budget tradeoff                   |
| Exact intervention  | Available                                         | Not available                               |

difference and a 16.3 ms prompt-heavy end-to-end tradeoff; the high-load regime showed a smaller late-time-to-first-token difference and a 23.3 ms prompt-heavy end-to-end tradeoff.

The direct FULL-versus-CHUNKED comparison did not reproduce the simulator’s qualitative reversal (Table 6). KV utilization and preemption metrics were unavailable in the native probe because its metric parser returned empty dictionaries. The real result therefore supports the narrower statement that concrete serving pressure can alter observable scheduling behavior and latency. It is not real-system causal confirmation of Eq. 3.

## 10. Discussion

### 10.1. When adaptation is justified

The evidence does not support an anti-adaptive-scheduling conclusion. It supports a staging rule. Under abundant resources, a complex portfolio may collapse to the same action as single best scheduler, making adaptation overhead difficult to justify. Under binding KV or active-sequence pressure, executable alternatives can appear. Some fresh disagreement states contain positive latency headroom, but opportunity prevalence and effect magnitude remain regime-specific.

This matters because scheduler diversity is not scheduler value. A system that produces many distinct actions may still optimize the wrong objective; a system with high conditional benefit may have too few disagreement states to justify a router. The decomposition in Eq. 4 is a practical diagnostic, not a promise of deployment gain. The measured 1.9958 ms mean headroom is therefore evidence that causal opportunity exists in the sampled disagreement-support regime; deciding whether that opportunity exceeds selector latency, engineering complexity, and operational risk is a separate deployment question.

### *10.2. Relation to modern adaptive systems*

LMetric, SOLA, Llumnix, MorphServe, Strata, Libra, FastServe, and related systems answer how to adapt serving behavior under dynamic resource and SLO conditions. ServeGen, BurstGPT, Azure traces, OpenTela, and longitudinal trace studies answer how to obtain or operate on realistic workload evidence. Our measurement asks when adaptation is unnecessary because canonical actions collapse, and when a local alternative is causally useful relative to a strong default. The questions are complementary; the present six-policy study does not claim to replace or exhaust modern schedulers, nor does the literature comparison substitute for a benchmark when making a direct performance claim.

### *10.3. Limitations*

The supported scope is the studied production-derived trace families, the specific pressure regimes tested, the controlled replay/intervention framework, and the bounded real-vLLM configuration. The six-policy portfolio is not the universe of 2026 schedulers, and different portfolios may expose different support. BurstGPT did not contribute a selected fresh causal regime. Fresh headroom is a mean-latency one-step oracle result, not a learned selector or production improvement; its mean effect is modest and may or may not exceed selector overhead. Fresh non-overlapping windows provide robustness within the trace families, not universal external validity. The real-vLLM validation uses one model, one RTX 5060 Ti GPU, vLLM 0.27.1, and one mechanism family, uses token-shape replay rather than exact private prompts, and supports the pressure/action mechanism only. It has no exact one-step real-system intervention, and the simulator-to-vLLM action mapping is imperfect. arrival-normalized weighted goodput saturation limits conclusions

under that objective. Finally, we make no claim over all LLM-serving workloads, GPU architectures, model sizes, scheduler portfolios, serving systems, or forms of resource pressure; no claim of learned predictability, universal adaptive benefit, universal scheduler superiority, or impossibility; and no claim that measured causal headroom automatically justifies deployment.

## 11. Reproducibility and Artifact Scope

The public artifact (the public repository, see Data and Code Availability) records trace provenance, frozen manifests, support grids, fresh non-overlap proofs, causal state and action manifests, source hashes, bootstrap seed 20260920, and the real-vLLM environment. The principal result packages are Phase A native replay, Phase B pressure mapping, fresh confirmatory causal latency, and real-vLLM pressure/action validation. Historical selector and joint-240 artifacts remain available but are not treated as evidence of learned deployment readiness.

## 12. Conclusion

Adaptive LLM serving should be evaluated in stages. Faithful production-derived replay can be action-null under abundant resources; resource pressure can create canonical executable alternatives; and fresh one-step replay can show that some alternatives contain positive local mean-latency headroom. In this study, the fresh confirmation found 590 beneficial states among 720 disagreement states. Mean oracle headroom was 1.9958 ms, with a positive corrected clustered confidence interval. The result is conditional and modest: the conditional benefit probability does not equate to overall deployment value.

Terminal arrival-normalized weighted goodput saturation demonstrates why objective choice matters, while bounded vLLM evidence provides real-system support for the pressure mechanism and observable action differentiation without claiming real causal headroom or deployment value. This performance-evaluation methodology shows that adaptation should be justified by measured opportunity and causal value; closed-loop and deployed value evaluation remains future work.

## **Declaration of generative AI and AI-assisted technologies in the manuscript preparation process**

During the preparation of this work, the author used ChatGPT by OpenAI, OpenAI Codex, and Google Gemini in order to assist with organization, language refinement, literature-search planning, software development, and workflow support. After using these tools, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

## **Declaration of Competing Interest**

The author declares that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Data and Code Availability

Code, derived research artifacts, experiment configurations, and reproducibility artifacts for this study are available at <https://github.com/SoroushVahidi/llm-serving-heuristic-evolution>. Raw third-party workload traces are obtained from their original upstream sources. The derived research data and reproducibility package have been permanently archived at DOI: <https://doi.org/10.5281/zenodo.22865294>.

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